

3. (a) Let us consider $P(x) = \sum_{n=0}^{\infty} a_n x^n$ (whenever exists). If $P(5)$ exists, will $P(4)$ exist? Justify.

(b) Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$.

i. Show that the series converges uniformly on every interval $[-r, r]$ with $0 \leq r < 1$.

ii. Justify whether the series also converges uniformly on the interval $[-1, 1]$. [1+3+2]

$$(a) P(x) = \sum_{n=0}^{\infty} a_n x^n$$

we know that if the series converges for $x = x_0$ ($x_0 \neq 0$) then it uniformly converges for $|x| \leq R < |x_0|$

$\Rightarrow$ if it exists for $x = 5$ then take $R = 4.5$ it converges (exists)

for $x \in [-4.5, 4.5]$

hence $P(4)$ exists as $4 \in [-4.5, 4.5]$

on $[-r, r]$ with $0 \leq r < 1$.

ii. it does not converge uniformly on the interval $[-1, 1]$

for $(-1, 1)$ it converges

to 0 as $n \rightarrow \infty$

$$(b) (i) \text{ series } P(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$$

$$\text{for } x=1 \quad P(1) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

we know that this converges.

also if $P(x)$ converges uniformly for $x = x_0$ then it converges for $|x| \leq r$ where $r < |x_0|$

here $x_0 = 1$

$\Rightarrow P(x)$ converges uniformly on $[-r, r]$

..... Rough Work

$$\begin{aligned} \|F(x_1, x_2) - F(y_1, y_2)\| &\leq \|(x_1, x_2) - (y_1, y_2)\| + (x_1 - y_1)^2 + (2x_2 - 2y_2)^2 \\ &\quad + 2(x_1 - y_1)(2x_2 - 2y_2) \\ &\leq \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} + \frac{2(x_1 - y_1)^2 + 5(x_2 - y_2)^2}{16} \\ &\quad + \frac{y_1^6}{16} (x_2 - y_2) \\ &\leq (x_1 - y_1)^2 + \frac{5}{16} (x_2 - y_2)^2 \\ &\quad + x_1^2 + y_1^2 - 2x_1 y_1 \\ &\quad + 2(x_1 - y_1)(x_2 - y_2)^6 \end{aligned}$$

4. (a) Let $f: M \rightarrow N$ be continuous and $\{x_n\}$ be a Cauchy sequence in M . Will $\{f(x_n)\}$ be Cauchy? Justify.
- (b) Let $f: M \rightarrow N$ be uniformly continuous and $\{x_n\}$ be a Cauchy sequence in M . Will $\{f(x_n)\}$ be Cauchy? Justify.
- (c) Suppose $f: (0,1) \rightarrow \mathbb{R}$ is continuous, and for each $n \in \mathbb{N}$, f is uniformly continuous on $[\frac{1}{n}, 1]$. Must f be uniformly continuous on $(0,1)$? Justify your answer.

[2+3+2]

(a) No. only if it is uniformly continuous.

take $f: (0,1] \rightarrow \mathbb{R}$

where $f(x) = \frac{1}{x}$

this function is continuous

take $\{x_n\} := \frac{1}{n}$

this is Cauchy in $(0,1]$ as $|\frac{1}{m} - \frac{1}{n}| \rightarrow 0$ as $m, n \rightarrow \infty$

but $f(x_n) = n$ is not Cauchy

as $|f(x_{n+1}) - f(x_n)|$

take $\epsilon = \frac{1}{2}$, $\forall N$

and $|f(x_{N+1}) - f(x_N)| = |N+1 - N| = 1 > \epsilon$

$\Rightarrow$ $\nexists$ no N s.t. $|f(x_m) - f(x_n)| < \epsilon$ $\forall m, n \geq N$

hence not Cauchy.

(b) Yes. (M, d) (N, ρ)

Let f be uniformly continuous

$\forall \epsilon > 0, \exists \delta_\epsilon > 0$ s.t.

$d(y, x) < \delta_\epsilon$

$\Rightarrow \rho(f(y), f(x)) < \epsilon$

$\forall x, y \in M$.

$\{x_n\}$ is Cauchy

$\forall \epsilon > 0 \exists N_\epsilon$ s.t. $d(x_n, x_m) < \epsilon$

$\forall n, m \geq N_\epsilon$.

Solⁿ: take $\epsilon > 0$,

$\exists \delta_\epsilon > 0$ s.t.

$d(y, x) < \delta_\epsilon$

$\Rightarrow \rho(f(y), f(x)) < \epsilon$

& for this δ_ϵ , $\exists N$ s.t.

$d(x_n, x_m) < \delta_\epsilon \forall n, m \geq N$

$\Rightarrow \rho(f(x_n), f(x_m)) < \epsilon$ $\forall n, m \geq N$

this is the definition of Cauchy sequence.

(c) No. take

the fn.

① $f(x) = \frac{1}{x}$ in $(0,1]$

it is continuous

② also f is uniformly continuous on $[\frac{1}{n}, 1] \forall n$

Proof: (as it is ptwise cont.)

$\exists \delta$ at $\frac{1}{n}$.

Let $\delta = \delta_{\frac{1}{n}}$.

at $\frac{1}{n} + \delta$ $|y - x| < \delta$.

implies

this δ_1 is smaller

than δ for all $x \in (\frac{1}{n}, 1]$

hence. $\left| \frac{1}{n+\delta} - \frac{1}{n} \right|$ decreases as n increases

but this is not uniformly continuous on $(0,1]$ the fn converges ptwise.

take $|f(n+\delta) - f(n)| < \epsilon$

$\Rightarrow \left| \frac{1}{n+\delta} - \frac{1}{n} \right| < \epsilon$ as n increases the δ allowed also increases

$\Rightarrow \delta_{\frac{1}{n}}$ can satisfy the

condition for all $n \in [\frac{1}{N}, 1]$

hence it is uniformly convergent on $[\frac{1}{N}, 1]$

but it is not uniformly convergent on $(0,1]$

Let say δ which is valid for all n . as $\delta \rightarrow 0$ and hence no δ can satisfy

..... Rough Work

$\rho(f(x_m), f(x_n)) < \epsilon$.

$\frac{1}{n} (x_1 + x_2 - y_1 - y_2)$,

$|f(y) - f(x)| < \epsilon$

$|f(n+\delta) - f(n)| < \epsilon$

$\left| \frac{1}{n+\delta} - \frac{1}{n} \right| < \epsilon$.